Sahaj Bhadja

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Summary

Software engineer with 2+ years building scalable microservices, full-stack applications, and applied ML systems. Experienced in Python, Java, C++, React, Node.js, REST APIs, relational databases, schema design, algorithms, testing, debugging, monitoring, and production deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python microservice APIs** and Azure SQL systems for 500+ users, optimizing indexed queries, schemas, and backend execution paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented 5,000+ monthly workflows through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving **monitoring**, recovery, maintainability, operational resilience, and systematic production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **React interfaces and Flask APIs** handling 10,000+ daily requests, implementing reusable workflows, authentication, progress states, and operational controls while systematically debugging production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, **AWS RDS** persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based **Decision Transformer** for offline reinforcement learning, improving long-horizon trajectory planning 18% in closed-loop simulation while developing end-to-end ML pipelines processing 100,000+ trajectory logs.
- Architected a spatial-temporal encoder integrating map geometry and agent context through **multi-head attention**, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable **Python services and REST integrations** using object-oriented patterns, AWS Glue, S3, SQL, filtering, pagination, and caching, improving production response times 30% and processing efficiency 68%.
- Maintained 20+ automated unit and integration suites with **Jenkins**, PyTest, and Mockito, accelerating defect investigation, release verification, and reliable deployments across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular **React and Node.js** features across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, processing 2 million+ Azure SQL records through documented software interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through **Jenkins and Azure**, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

Skills

Languages & Fundamentals: Python, Java, C++, JavaScript, TypeScript, SQL, object-oriented programming, data structures, algorithms, operating systems, concurrency, multithreading
Frontend & Backend: React, Next.js, Node.js, Flask, FastAPI, REST APIs, microservices, WebSockets, authentication, role-based access, responsive web applications, distributed workflows
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, AWS RDS, AWS, Azure, S3, relational databases, schema design, query optimization, caching
ML & Engineering: PyTorch, machine learning, artificial intelligence, model evaluation, Git, Jenkins, CI/CD, Docker, Kubernetes, unit testing, integration testing, debugging, monitoring, profiling, root-cause analysis, documentation, Agile

Projects

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a scalable platform supporting 50,000+ users and one million transactions, integrating **microservice APIs**, real-time WebSocket synchronization, MongoDB persistence, Kubernetes deployment, production monitoring, and automated release workflows.

Big-Brain | Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*
- Built an **AI-powered document platform** using LLaMA-2, FAISS, PostgreSQL, and REST APIs, enabling contextual search and collaborative analysis across 50,000+ documents while reducing review time by 60%.

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms, Operating Systems*
- Built a **C++ simulator** implementing six scheduling algorithms with deterministic workloads, edge-case validation, Gantt visualization, and performance testing, reducing average turnaround time 25% across 1,000+ process configurations.